

lgbm_3class_xfn5d_30f — Model Card

LightGBM

3-Class

Locked 2026-04-22

TD Bank directional signal detector — predicts 5-day excess return vs XFN sector ETF

60%

Active sign accuracy

0.566

Dir AUC (random = 0.50)

94%

Active coverage

30

Features

Problem Framing

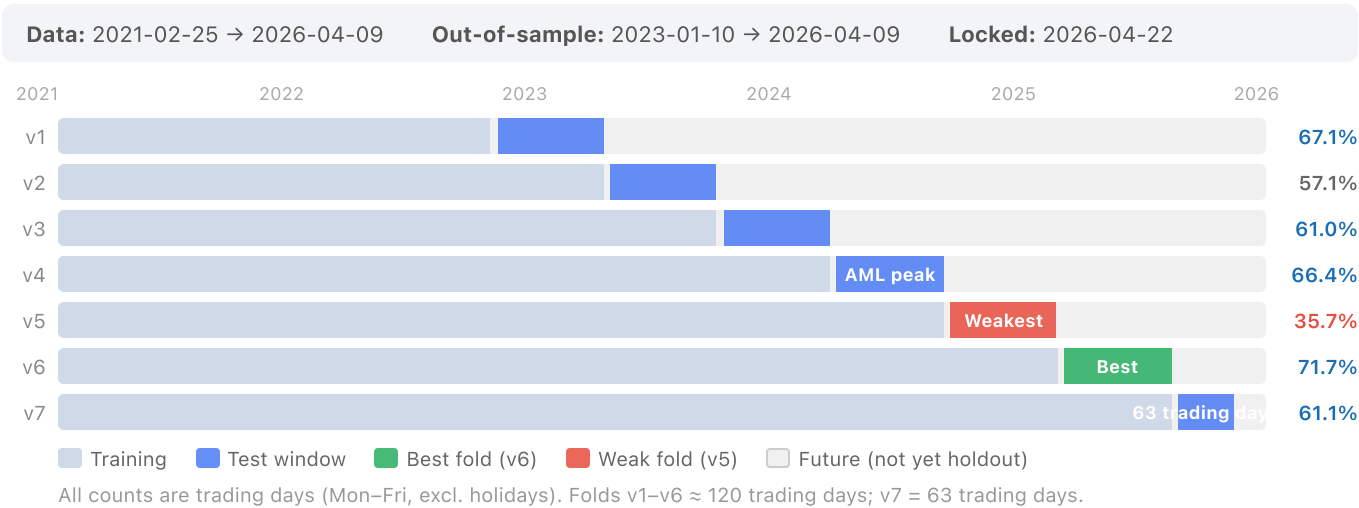
Given all information available on day d , the model outputs one of three signals for the next 5 trading days.

Signal	Label	Condition	Interpretation
Outperform	+1	Excess return > +0.30%	Take long position vs sector
Neutral	0	Excess return within $\pm 0.30\%$	No actionable signal
Underperform	-1	Excess return < -0.30%	Reduce / hedge vs sector

Training & Holdout Timeline

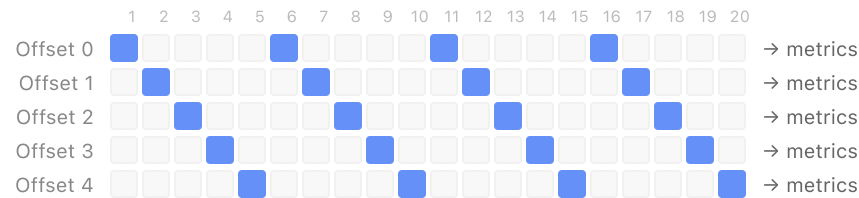
5-day gap between each train end and test start

Expanding-window walk-forward — each fold retrains on all history up to its cut-off. Score at right is fold active sign accuracy. All counts are **trading days**.



Evaluation: Stride-5 Offset Averaging

Predictions are made every day, but 5-day returns overlap — counting every row would repeat the same return 5 times. Stride-5 averaging selects one non-overlapping sequence per offset, then averages the 5 resulting metrics.



Average 5 metric values = reported fold score. Equivalent to treating each 5-day return as one independent observation.

Per-Fold Performance

Row highlight: **blue** = AML peak (v4), **amber** = weakest fold (v5), **green** = best fold (v6). Dir AUC coloured: **green** ≥0.60, **blue** 0.50–0.60, **red** <0.50 (below random).

Fold	Train window	Test window	Days	Mean acc	Macro F1	Active sign acc	Dir AUC	Coverage	Majority BL	Momentum BL
v1	2021-02-25 → 2022-12-30	2023-01-10 → 2023-06-30	121	50.4%	0.385	67.1%	0.622	95.9%	39.7%	39.0%
v2	2021-02-25 → 2023-06-30	2023-07-11 → 2023-12-29	119	43.7%	0.355	57.1%	0.574	94.2%	41.1%	37.0%
v3	2021-02-25 → 2023-12-29	2024-01-09 → 2024-06-28	121	44.7%	0.305	61.0%	0.543	90.9%	23.2%	39.6%
v4	2021-02-25 → 2024-06-28	2024-07-09 → 2024-12-31	121	55.3%	0.437	66.4%	0.643	95.9%	42.2%	35.6%
v5	2021-02-25 → 2024-12-31	2025-01-09 → 2025-06-30	120	29.2%	0.209	35.7%	0.329	93.3%	26.7%	49.2%
v6	2021-02-25 → 2025-06-30	2025-07-09 → 2025-12-31	121	55.3%	0.482	71.7%	0.709	93.4%	45.4%	30.6%
v7	2021-02-25 → 2025-12-31	2026-01-09 → 2026-04-09	63	49.4%	0.338	61.1%	0.545	96.7%	51.2%	25.5%
7-fold mean				46.8%	0.359	60.0%	0.566	94.3%	38.5%	36.6%

7-Fold Average vs Baselines

Majority and momentum are the most relevant benchmarks for a pension mandate.

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Model

60%

Active sign acc

Dir AUC: 0.566

Coverage: 94% | F1: 0.36

Majority

Baseline

38.5%

Active sign acc

Coverage: 100%

Always predict training-set majority class

Momentum

Baseline

39.0%

Active sign acc

Coverage: 79%

Predict continuation of TD vs XFN 5d return

Always-Long

Reference

43%

Active sign acc

Coverage: 100%

Not viable for pension — no drawdown protection

+21pp lift over majority and momentum on active sign accuracy · Dir AUC 0.566 vs 0.50 random

The model is directionally correct 60% of the time it takes a position. Dir AUC confirms this threshold-free: the model's probability scores rank outperformers above underperformers 56.6% of the time, vs 50% for a random model. It does not predict magnitude.

Feature Set (30)

NLP event features are computed from LLM-scored earnings call transcripts and projected to daily grain via exponential decay anchored to days_since_call (half-life 20 days).

Market / Price8

td_return_5dtd_return_20dtd_volatility_20dtd_vs_xfn_5dtd_vs_tsx_5dtd_corr_pv_20dtd_volume_change_20dtd_dist_52w_high

Rates / Macro2

yield_curve_slopeyield_10y_level

Global Risk4

vix_volatility_20ddxy_levelgold_levelfx_usdcad_level

News Flow3

news_sent_mean_30dnews_count_30ddays_since_last_news

Timing2

days_since_callis_earnings_week

NLP Event (decayed)11

evt_ceo_toneevt_cfo_toneevt_framing_gapevt_aml_pressureevt_aml_shiftevt_guidance_strengthevt_guidance_shiftevt_macro_topicevt_topic_entropyevt_news_toneevt_news_flow

Limitations

#	Limitation	Detail
1	Single stock	Trained only on TD Bank — signals are specific to TD's dynamics and earnings cadence
2	Directional only	Does not predict magnitude, only direction relative to XFN sector
3	AML feature timing	evt_aml_pressure/shift included with domain knowledge; documented in experiment log
4	Data dependency	Requires daily feature pipeline (model_features_daily.parquet) and price data